

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Maddirala Gopi Krishna Reddy 192372055(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

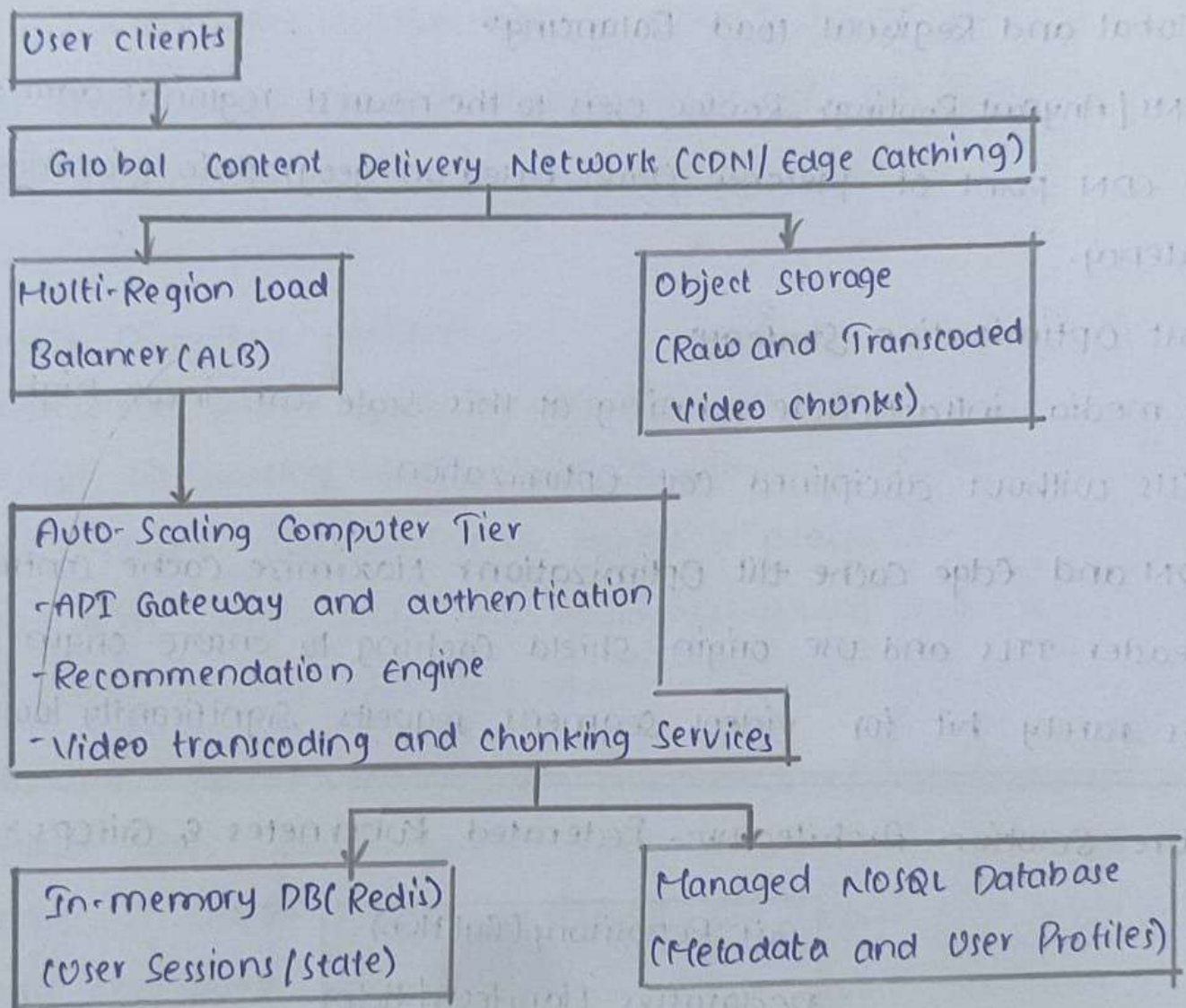
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. Cloud Architecture design:

To support millions of Concurrent video streams with low latency, the System uses a decoupled microservices-based cloud architecture.



Key architectural Concepts:

Storage tier: Raw video files are uploaded directly to Scalable Object Storage. Transcoded video segments are stored in partitioned, highly durable buckets.

Ingestion and Transcoding: Serverless functions or auto-scaled compute clusters convert uploaded videos into multiple bitrate resolutions broken into 2-6 second chunks.

Ensuring Smooth Streaming: Load Balancing and Auto-Scaling:

Handling millions of concurrent requests requires a resilient distribution and scaling strategy.

Global and Regional load Balancing:

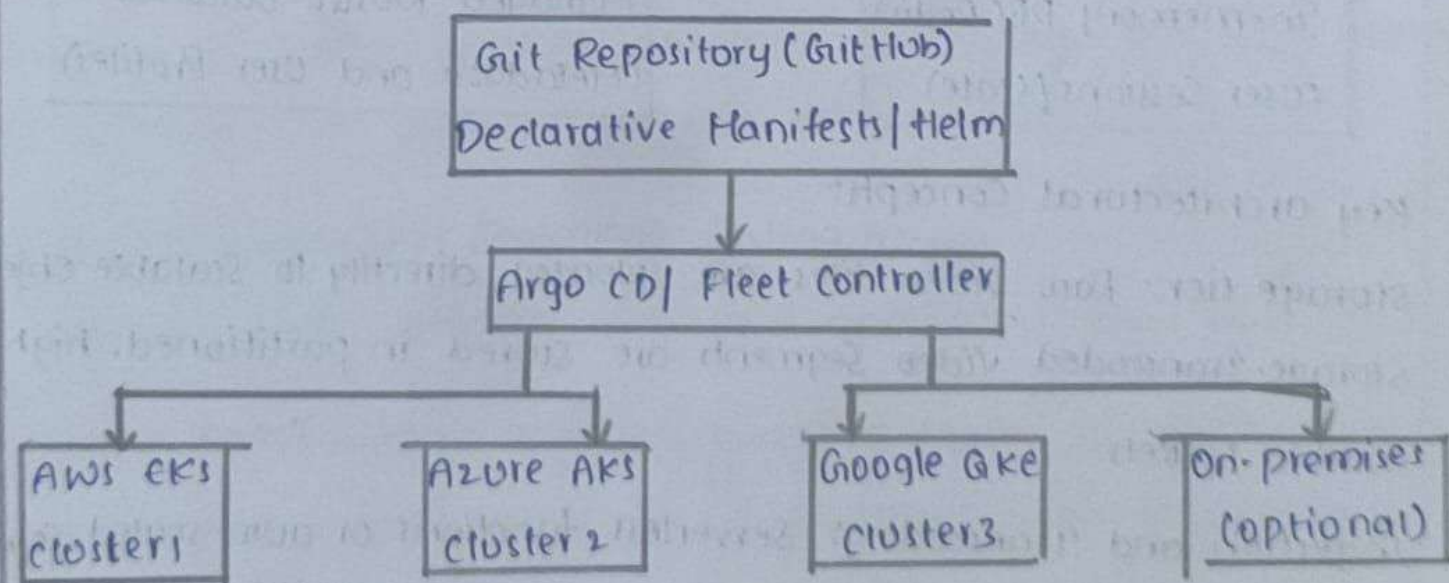
DNS / Anycast Routing: Routes users to the nearest regional data center or CDN point of presence (PoP) based on geographic proximity and latency.

Cost Optimization Strategy:

A media infrastructure running at this scale can incur high cloud bills without disciplined cost optimization.

CDN and Edge Cache Hit Optimization: Maximize cache control header TTLs and use origin shield caching to ensure origin servers are rarely hit for video segment requests significantly lower.

2. Core Solution Architecture: Federated Kubernetes & GitOps



Key architecture components:

Unified Control plane: Use multi-cloud Kubernetes platforms such as Google Distributed cloud or Azure Arc-enabled Kubernetes to

* Preventing and Rolling Back Accidental Deletion:

Object Versioning: Retains prior states when a file is overwritten or deleted. A standard delete operation merely attaches a "Delete Marker" rather than destroying the data. Reverting deletion requires dropping the marker.

MFA delete and Object lock: Enforces Multi Factor Authentication for permanent purge requests and locks Compliance backups under write-once-read-many Policies, neutralizing malicious insider threats and accidental script executions

Recovery Metric	Target Objective	Technical Strategy
Recovery point Objective (RPO) (Maximum tolerable data loss)	<15 seconds for metadata Near 0 for video files Near 0 for video files	Continuous transactional write log streaming to Secondary Storage Combined with asynchronous Cross-Region Replication
Recovery Time Objective (RTO) (Maximum acceptable downtime)	<15 minutes	Active passive warm Standby model with automated DNS failover, standby database promotions take seconds.

Benefits and Risks

Area	Benefits	Risk & Migration Strategies
Vendor-Lockin	Complete cloud neutrality applications can easily shift workloads if pricing or SLA policies change	Risk: Over abstraction leads to using the low-cost common denominator of cloud services Migration: Standardize runtime level while using managed services where appropriate.

3. To recover from accidental data deletion and regional cloud outages and elearning platform requires a multi-layer disaster recovery architecture on automated, cloud-native storage, continuous replication and immutable backups.

Solution Architecture: cloud native DR Engine

[Primary Region]

[Secondary/ DR Region]

Object Storage (Video/Documents)

- Object versioning (enabled)
- Object lock (WORM/Compliance)

Object Storage (DR Replica)

- Object lock (Immutable Copy)
- MFA Delete policy Enforcement

Managed Transactional Database

- Continuous point-in-time (PITR)
- Automated hourly Snapshots

Read Replica/ Standby database

- Automated Failover Promotion
- Daily Encrypted offsite

Connect, secure and govern clusters across AWS, GCP and Azure under a single management plane.

GitOps Deployment Strategy Maintain application stages declaratively in Git repositories. GitOps agents running inside each target cluster pull configuration changes directly from Git. This guarantees that every environment stays strictly identical regardless of the underlying cloud provider.

Service Discovery **Scaling**

• **Multi cloud Service Discovery and Networking**

Direct IP routing across cloud providers is challenging. The recommended approach is to deploy a Global Service Mesh integrated with external DNS and multi-cloud Global Server Load Balancer like Cloudflare or AWS Route 53.

Cross-cluster Service Mesh: Istio or Consul creates a flattened virtual network across clouds using encryption IPsec or WireGuard mTLS tunnels. A service running on GCP can resolve and route directly to an endpoint running on AWS without public exposure.

2. **Dynamic workload Scaling** Scaling across multiple clouds

requires both Compute-level and application-level metrics.

• **cluster AutoScaling** Use cloud native cluster autoscalers or

Karpenter on each respective cloud provider to dynamically

add worker nodes when pending pods require additional

compute capacity.

4. Hybrid Architecture & Secure Connectivity Design

On-premises Data Center

[Sensitive Databases and Regulated Legacy Apps]



[Hardware Security Module (HSM)] - [App Gateway]

Direct Dedicated Link (e.g., AWS Direct/Azure ExpressRoute)
+ MACsec (Layer 2 Encryption) or IPsec Tunnel (AES-256)

Public cloud environment

[Transit Gateway / Virtual Hub]

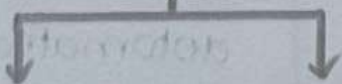
[Non-Sensitive Public Web] [Analytics & processing]

Identity Management & Encryption Framework

[Identity Governance]

[Enterprise Idp]

[Central ID / Ping Federate]



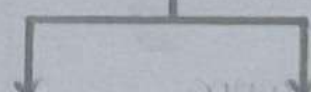
[Role-Based] [SPIFFE / SPIRE]

Access Control Workload Ident

[Data protection]

[FIPS 140-3 HSM]

[On-prem / cloud]



[TLS 1.3]

In Transit

[AES-256-GCM]

At Rest

Key Operational Challenges in Government Hybrid Environments

Challenge Area	Description	Mitigation Strategy
Audit & Compliance Fragmentation	Difficulty generating unified compliance reports across on-premises and multi-cloud boundaries	Deploy Continuous Compliance platforms AWS Security Hub, Azure Policy feeding into a SIEM system
Identity and Secret Drift	Long-lived credentials or inconsistent revocation policies across hybrid systems create privilege escalation	Implement dynamic Secret engines with short-lived tokens and continuous automate.

5. Multi-Region Strategy (Active-Active)

Global Load Balancing: Use Anycast DNS routing to direct users to the nearest regional datacenter.

Stateless App tier: Containerized app clusters (EKS/GKE) across multiple regions so any user request can load on any region.

2. Data Replication and Failover:

Databases: Use globally distributed database (Aurora Global)

DynamoDB Global Tables, Spanner) with asynchronous Sub-Second Cross-region replication.

Storage: Enable Cross-Region Replication (CRR) on Object Storage buckets (S3/GCS)

Automated Failover: Global health checks probe endpoints every 5-10 seconds. If a region fails, traffic automatically reroutes to the healthy region with zero or minimal downtime.

3. Monitoring and Alerting:

Track Replication lag: Alert immediately if database replication lag exceeds 2 seconds

Synthetic Edge Monitoring: Run automated scripts from edge nodes to test real user logins/checkout flows every minute.

Chaos Testing: Run simulated region-outage drills to ensure failover routines work reliably.